



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

KARAN DAVE, 26070122507, CSE B3, 2025-2029

ASSIGNMENT No: 12

TITLE: Write a Java program to create and use user-defined packages.

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Theory

Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Largescale software systems extensively use packages.



Exercise

1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

```
1 class Student {
2     String name;
3     int rollNo;
4     String course;
5     Student(String name, int rollNo, String course) {
6         this.name = name;
7         this.rollNo = rollNo;
8         this.course = course;
9     }
10    void displayStudent() {
11        System.out.println("Student Details:");
12        System.out.println("Name: " + name);
13        System.out.println("Roll No: " + rollNo);
14        System.out.println("Course: " + course);
15    }
16 }
17 class Faculty {
18     String name;
19     int facultyId;
20     String department;
21     Faculty(String name, int facultyId, String department) {
22         this.name = name;
23         this.facultyId = facultyId;
24         this.department = department;
25     }
26    void displayFaculty() {
27        System.out.println("Faculty Details:");
28        System.out.println("Name: " + name);
29        System.out.println("Faculty ID: " + facultyId);
30        System.out.println("Department: " + department);
31    }
32 }
33 public class Main {
34     public static void main(String[] args) {
35         Student student = new Student(
36             "Karan", 101, "Computer Engineering"
37         );
38         Faculty faculty = new Faculty(
39             "Sujay", 501, "Civil Engineering"
40         );
41         System.out.println("COLLEGE MANAGEMENT SYSTEM");
42         System.out.println();
43         student.displayStudent();
44         System.out.println();
45         faculty.displayFaculty();
46     }
47 }
48 }
```

<terminated> Main (3) [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (17-Aug-2026, 12:03:30a)
COLLEGE MANAGEMENT SYSTEM

Student Details:
Name: Karan
Roll No: 101
Course: Computer Engineering

<terminated> Main (3) [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (17-Aug-2026, 12:03:30a)
COLLEGE MANAGEMENT SYSTEM

Student Details:
Name: Karan
Roll No: 101
Course: Computer Engineering



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COLLEGE MANAGEMENT SYSTEM

Student Details:

Name: Karan

Roll No: 101

Course: Computer Engineering

Faculty Details:

Name: Sujay

Faculty ID: 501

Department: Civil Engineering



2. Create a Java program by developing a user-defined package named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

```
Program2.java
1 class Library {
2     int bookId;
3     String title;
4     String author;
5     double price;
6     Library(int bookId, String title, String author, double price) {
7         this.bookId = bookId;
8         this.title = title;
9         this.author = author;
10        this.price = price;
11    }
12    void displayBook() {
13        System.out.println("Book ID: " + bookId);
14        System.out.println("Title: " + title);
15        System.out.println("Author: " + author);
16        System.out.println("Price: Rs. " + price);
17    }
18 }
19 public class Program2 {
20     public static void main(String[] args) {
21         Library book = new Library(
22             101,
23             "The Alchemist",
24             "Karan Dave",
25             3999.00
26         );
27         System.out.println("LIBRARY BOOK DETAILS");
28         System.out.println();
29         book.displayBook();
30     }
31 }
```

Problems @ Javadoc Declaration Console Properties
terminated> Main (3) [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (17-Aug-2026, 12:03:30 a
OI | FGF MANAGMFMT SYSTFM
Writable Smart Insert 32 : 1 : 865
ENG IN 60% 00:08 17-08-2026

LIBRARY BOOK DETAILS

Book ID: 101

Title: The Alchemist

Author: Karan Dave

Price: Rs. 3999.0